

Lab #:

Date:

Title: Calcium Carbonate content of Egg shells

Topic: Back Titration

Skill(s):

Aim: To determine the Calcium Carbonate content in Egg Shells using Back titration.

Apparatus/Materials:

Diagram:

Method:

1. Obtain an egg and crack it to remove the inner contents and secure the shell.
2. Remove the inner membrane of the egg shell and place the remaining portion in an oven for 10 minutes until it is sufficiently dried.
3. Place the dried egg shell in a mortar and use a pestle to crush and grind the egg shell to powder.
4. Weigh exactly 0.50 g of the egg shell in an Erlenmeyer flask.
5. Use a measuring cylinder to measure 60 mL of the standard HCl solution into the Erlenmeyer flask that contains the egg shell and allow the reaction to proceed for 2-3 minutes while swirling.
6. After 2-3 minutes of adding the HCl to the egg shell, heat the mixture in a water bath for 10 minutes. Cool it down and add 3 drops of phenolphthalein indicator.
7. Titrate the mixture with the NaOH solution until the color changes to pink.
8. Repeat steps (4) to (7).
9. Calculate the percentage of Calcium carbonate in the egg shell in each replicate.

10. Calculate the mean percentage of calcium carbonate in the egg shell.

Results:

TRIAL	1	2	3
Final burette reading (mL)			
Initial burette reading (mL)			
Volume of NaOH used (mL)			

Calculations:

1. Write a chemical equation to represent the reaction of the analyte with the excess standard reagent.
2. Write a chemical equation to represent the reaction of the left-over standard solution with the second standard solution.
3. Calculate the initial moles of HCl added to the egg shells.
4. Calculate the moles of HCl remaining in the flask after reaction with analyte.
5. Calculate the moles of HCl consumed by the analyte.
6. Calculate the mass of CaCO_3 in the analyte.
7. Determine the Percentage composition of CaCO_3 in the analyte.

Discussion:

1. Why use a back titration for CaCO_3 instead of direct approach?
2. How would an incomplete removal of the eggshell membrane affect the calculated % CaCO_3 ? Explain direction and why.

3. If the measured % CaCO_3 is much lower than literature values (~90–98 %), list two plausible experimental errors that could cause this and describe a test that would determine which one happened.
4. How does using a stronger HCl (say 1.0 M) change experimental convenience and potential errors? Explain pros and cons.

Precautions:

Sources of Error:

Assumption(s):

Conclusion: